

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF
ENGINEERING)** Department of Computer Science and
Engineering

ASSESSMENT TOOL 2 – Industry Problem Solving Task

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO4: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Industry Problem Solving Task
Weightage:	20% of CIA	Total Marks:	100
CO4 Statement:	Solve optimization problems using dynamic programming and greedy strategies.		
Student Name:	J Venkata Varun	Reg. No.:	192372227
Section / Batch:	AI	Date:	26-08-26

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues.
- Provide a thorough analysis with validated results and performance evaluation.

Question Type	Focus Area	Questions
Industry Problem Solving Task	Focus on Solving optimization problems using dynamic programming and greedy strategies	Q1

SECTION A – Industry Problem Solving Task

A multi-specialty hospital must allocate limited ICU beds and medical staff among critical patients. Analyze how optimization algorithms can improve resource utilization. Identify constraints such as patient priority, emergency arrivals, and staff availability. Design a schedule model using efficient algorithms. Discuss feasibility in real-time healthcare systems. Evaluate performance based on response time and a scheduling utilization. Interpret results in improving patient care efficiency.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION**Industry Problem Solving Task**

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
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Understanding of Industry Problem	20%				
Application of Engineering Concepts	25%				
Solution Design	25%				
Use of Tools	15%				
Analysis	10%				
Professionalism	5%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



SIMATS

ENGINEERING

SIMATS Online Programming Lab

DAA PROGRAMMING

NARRAVULA NAVYA SRI

TRIZAGADDERA

QUESTIONS

Q1
Smart Healthcare Resource Allocation (Optimization + Scheduling)
100 marks
1/1 answered

Q1 Smart Healthcare Resource Allocation (Optimization + Scheduling) 100 marks

A multi-specialty hospital must allocate limited ICU beds and medical staff among critical patients. Analyze how optimization algorithms can improve resource utilization. Identify constraints such as patient priority, emergency arrivals, and staff availability. Design a scheduling model using efficient algorithms. Discuss feasibility in real-time healthcare systems. Evaluate performance based on response time and utilization. Interpret results in improving patient care efficiency.

Your Code

```
1 import heapq
2 class Patient:
3     def __init__(self, patient_id, priority, arrival_time):
4         self.patient_id = patient_id
5         self.priority = priority
6         self.arrival_time = arrival_time
7     def __lt__(self, other):
8         if self.priority == other.priority:
9             return self.arrival_time < other.arrival_time
10        return self.priority < other.priority
11 class HealthcareResourceManager:
12     def __init__(self, total_icu_beds, total_staff):
13         self.available_beds = total_icu_beds
14         self.available_staff = total_staff
15         self.patient_queue = []
16         self.allocated_resources = {}
17     def add_patient(self, patient):
18         heapq.heappush(self.patient_queue, patient)
19         print(f"Patient {patient.patient_id} added to queue (Priority: {patient.priority})")
20     def allocate_resources(self):
21         print(f"--- Starting Resource Allocation ---")
22         while self.patient_queue and self.available_beds > 0 and self.available_staff > 0:
23             patient = heapq.heappop(self.patient_queue)
24             self.available_beds -= 1
25             self.available_staff -= 1
26             self.allocated_resources[patient.patient_id] = {"beds": 1, "staff": 1}
27             print(f"Allocated resources to Patient {patient.patient_id}: {self.allocated_resources[patient.patient_id]}")
28             print(f"Remaining Beds: {self.available_beds}, Remaining Staff: {self.available_staff}")
29         if self.patient_queue:
30             print(f"Resource Shortage! (len(self.patient_queue) > 0)")
31         else:
32             print(f"All patients have been processed successfully.")
33         return self.allocated_resources
```

Input

```
3 3
101 3 0
102 1 1
103 2 2
```

Output

```
Patient 101 added to queue
(Priority: 3)
Patient 102 added to queue
(Priority: 1)
Patient 103 added to queue
(Priority: 2)
```